Emanuele Plebani

Palo Alto, CA | banus80@gmail.com | +1 (317) 645-5666
banus.github.io | GitHub: Banus | US Permanent Resident

Introduction

Machine learning engineer with industry and academic experience specializing in natural language processing, deep learning, and probabilistic models. I have a proven track record in developing and maintaining large-scale software projects and experience in delivering strategic leadership, mentoring teams, conducting in-depth research, and driving development initiatives. I have consistently demonstrated the ability to manage demanding workloads and release key features within strict deadlines.

Skills

Languages:	Python, C/C++, Javascript, SQL, Perl, Java, Haskell
Tools:	Tensorflow, PyTorch, Transformers, Scikit-Learn, GIT, Docker
Others:	language models, natural language processing, machine learning, anomaly detection, statistical modeling, data visualization and analysis, software engineering

Professional Experience

Jan 2026 – Present

Senior Deep Learning Engineer at Imagry, San Jose, CA

- Development of object detection and mapping models for autonomous driving
- Data preparation, cleaning and filtering from driving sequences
- Porting and testing models in C++ for the driving firmware

Dec 2024 – Jan 2026

Software Development Engineer II at Amazon, Santa Clara, CA

- Development of new AI-based shopping features for Alexa+, including Shopping Essentials, tracking deals on products and Autobuy based on price conditions
- Contribute to the design and implementation of LLM-based service APIs and data flows, mapping service dependencies, designing integration and load tests
- Maintain and improve existing Alexa Shopping features, including deal search and notifications, and integrating them in the new AI-based flows
- Monitoring and improving the service infrastructure through Infrastructure as Code (IaC) on AWS, fixing configuration issues, and improving code quality

Aug 2022 – May 2022

Research scientist intern at Meta, Redmond (WA)

- Developing an uncertainty-aware neural network model for eye tracking in the Aria project, implementing different algorithms to estimate prediction intervals, such as pinball loss, and quality driven (QD).
- Developing out-of-distribution approaches to identify anomalous training sequences, implementing fast ensemble methods.

May 2024 – Jan 2020

Research Assistant at IUPUI, Indianapolis | Under Prof. M. Dundar

- Automated tagging and summarization of medical notes by fine-tuning recurrent neural networks and Transformer language models in pytorch
- Mining a large language model with entailment and embedding similarity to augment a dataset of short messages
- Data imputation and preprocessing, training of XGBoost models on tabular data in R
- Classification of mineral signatures in hyperspectral images from the CRISM Mars experiment using Bayesian models, deep models, SVM, random forests
- Segmentation of different cell and nerve structures in neurohistological TEM images using U-Net
- Implementation of semi-supervised segmentation methods based on CNN and Transformers
- Implementation of unsupervised models for fine-grained class discovery
- Investigating Bayesian models for problems with imbalanced classes and zero-shot learning in segmentation and hyperspectral classification

Jan 2020 – Jun 2014 | **Advanced Research Engineer** at STMicroelectronics, Milan

- Initial prototype, design, development, and scaling of the Cube.AI back-end, a tool to port neural networks from different deep learning toolboxes to STMicroelectronics platforms
- Technical lead and project management in the Cube.AI project, involving up to 30 people and 5 locations across Italy and France
- Managing testing, build automation and integration, documentation, code review, and code standards for machine learning projects.
- REST API design, setup, and documentation (OpenAPI) of a container-based cloud-based web service to demo the Cube.AI app internally and to customers.
- Optimizing inference and memory of action recognition and object detection neural networks.
- Development of deep learning solutions for sensor data classification and activity recognition on microcontrollers, handling the full machine learning pipeline, including data collection, cleaning, incremental training, and field evaluation.
- Design and training of binary-valued neural networks for activity recognition
- Development and implementation of computer vision algorithms for visual search and object detection, with automated evaluation and result visualization

May 2014 – Oct 2011 | **Research Assistant** at Politecnico di Milano, Milan

- Development and implementation of content-based image retrieval algorithms for the MPEG CDVS standard
- Development of visual search algorithms with improved localization, multiple object search, and continuous tracking
- Person in charge of the deliverables assigned to Politecnico di Milano in the European Project ASTUTE ARTEMIS
- Advised theses on visual odometry using the trifocal tensor on omnidirectional images and robust plane detection in point clouds

Education

May 2024 PhD in Computer Science, **Purdue University**
GPA: 4/4 | Advisor: Prof. Murat M. Dundar

Jul 2011 Master in Computer Science Engineering, **Politecnico di Milano**
100/100 *First Class Honors* | Major: Engineering of Computing Systems

Selected Publications

[1] **E. Plebani**, N. Biscola, L. Havton, B. Rajwa *et al.*, "High-throughput segmentation of unmyelinated axons by deep learning", Scientific Reports 12, no. 1 (2022): 1198.

[2] **E. Plebani**, B. L. Ehlmann, E. K. Leask, V. K. Fox, and M. M. Dundar, "A Machine Learning Toolkit for CRISM Image Analysis", Icarus 376 (2022): 114849.

[3] M. Paracchini, E. Plebani, M.B. Iche, D. Pau and M. Marcon, "Embedded real-time visual search with visual distance estimation." Image Analysis and Processing-ICIAP September 11-15, 2017.

[4] M. Paracchini, A. Schepis, M. Marcon, M. Falchetto, **E. Plebani**, D. Pau. "Accurate omnidirectional multi-camera embedded structure from motion." RTSI 2016, pp. 1-6.

[5] A. De Vita, G.D. Licciardo, L. Di Benedetto, D. Pau, **E. Plebani** and A. Bosco, "Low-power design of a gravity rotation module for HAR systems based on inertial sensors." ASAP, 2018, pp. 1-4.

[6] A. Nicosia, D. Pau, D. Giacalone, **E. Plebani**, A. Bosco, A. Iacchetti, "Efficient light harvesting for accurate neural classification of human activities", ICCE, 2018, pp. 1-

[7] P. Karimi, **E. Plebani**, D. Bolchini. "Textflow: Screenless access to non-visual smart messaging." In 26th International Conference on Intelligent User Interfaces, pp. 186-196. 2021.

[8] D. Pau, **E. Plebani**, F. De Ambroggi, *et al.*, "Method, system, and computer program product to employ a multi-layered neural network for classification", US Patent 11,537,840

[9] R. Varenne, J.M. Delorme, **E. Plebani**, D. Pau, V. Tomaselli, "Intelligent recognition of TCP intrusions for embedded micro-controllers", New Trends in Image Analysis and Processing – ICIAP, pp. 361-373, 2019

Awards

Apr 2024 Gersting's Award for an Outstanding Graduate Student - **IUPUI**

Feb 2014 Prize for Best Demo at the GITTI MMSP Conference (Italian Chapter IEEE Signal Processing)